

Analysis and Prediction of Forest Cover Types

DSAA2011 Machine Learning Course Project

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581,012 Total Samples

A massive repository of georeferenced observations covering 30x30 meter cells in the Roosevelt National Forest, Colorado.

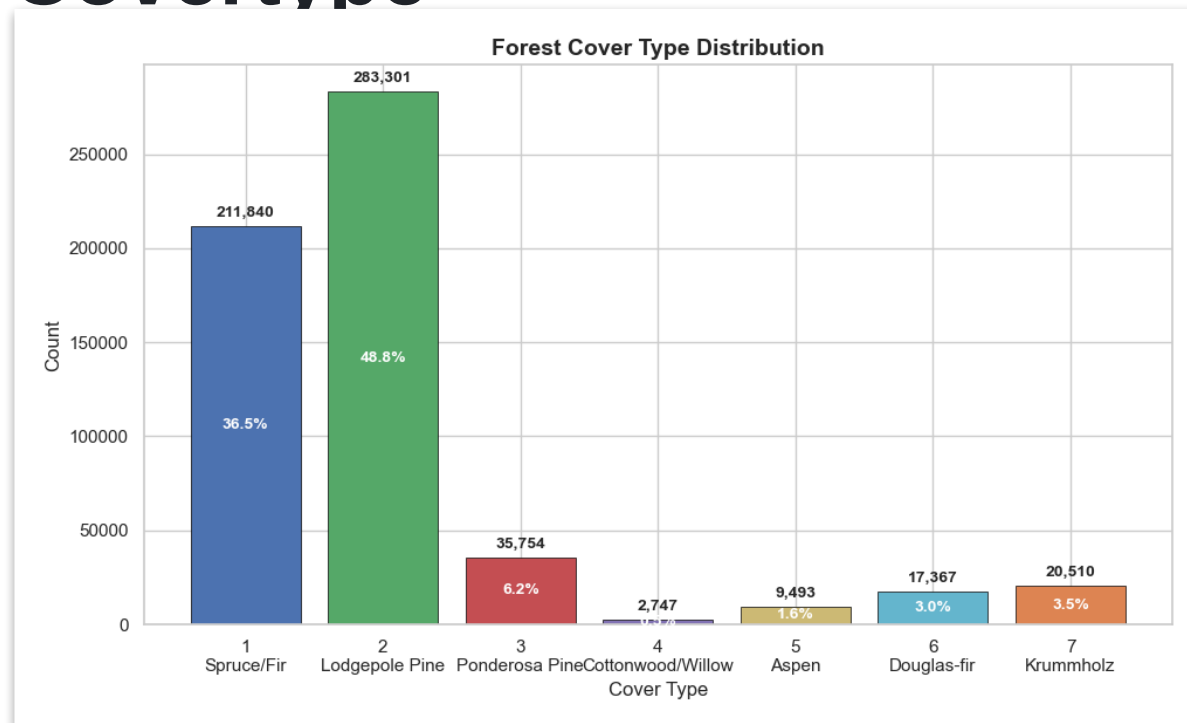
54 Cartographic Features

Includes 10 quantitative metrics (elevation, slope), 4 binary wilderness area flags, and 40 binary soil type indicators.

7 Forest Cover Classes

Classifies forest cover based on dominant tree species, ranging from Spruce/Fir (Class 1) to Krummholz (Class 7).

01 Dataset: UCI Forest Covertype



Key Finding: Severe Class Imbalance

Classes 1 and 2 account for 85.3% of all observations. Cottonwood/Willow is only 0.5%, so model selection emphasizes macro-F1 rather than accuracy alone. The supervised modeling sample contains 97,950 observations (≈17% of the full dataset).

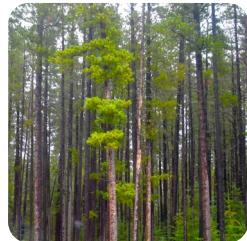
Overview of the Seven Forest Cover Types

Forest cover types are fundamental indicators for ecological monitoring and land use analysis. This page visually presents seven representative forest vegetation types in North America through typical imagery, from high-altitude coniferous forests to riparian broadleaf forests.



01 Spruce/Fir

Cold-hardy evergreen conifers with dense foliage, adapted to cold high-altitude climates, and are dominant species in the boreal forest.



02 Lodgepole Pine

Tall, straight trunks, often densely distributed, and one of the pioneer species for post-fire forest recovery in western North America.



03 Ponderosa Pine

Distinctive orange-yellow scaly bark, drought-tolerant, mainly distributed in warm, dry low-altitude mountainous areas.



04 Cottonwood/Willow Typical riparian deciduous broadleaf forests, fast-growing, highly dependent on water, with unique winter branch morphology.



05 Aspen

Smooth white trunks, golden leaves in autumn, often forming spectacular pure stands. As a fast-growing species, it plays an important role in forest succession.



06 Douglas-fir

One of the tallest tree species in the world, with a straight shape and excellent wood quality. It is not only an important commercial timber but also a key component of forest ecosystems.



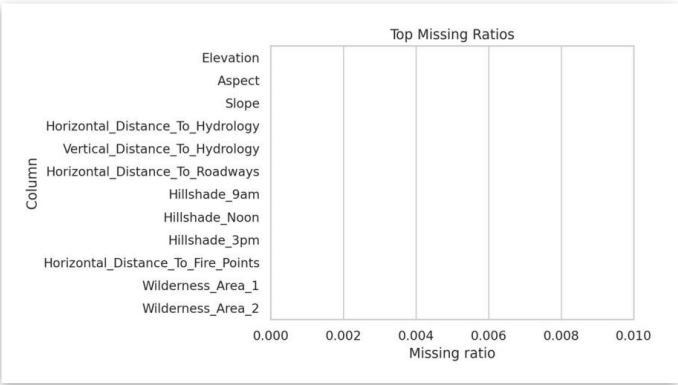
07 Krummholz

Growing above the alpine tree line, shaped into low, curved shrubs by strong winds and severe cold, it is a unique symbol of high-altitude ecosystems.

02 Data Preprocessing

Missing Value Audit

No null values detected across all 54 features



Feature Engineering Summary Matrix

Leakage control: StandardScaler is fitted only on the relevant analysis/training subset; all binary indicators remain 0/1.

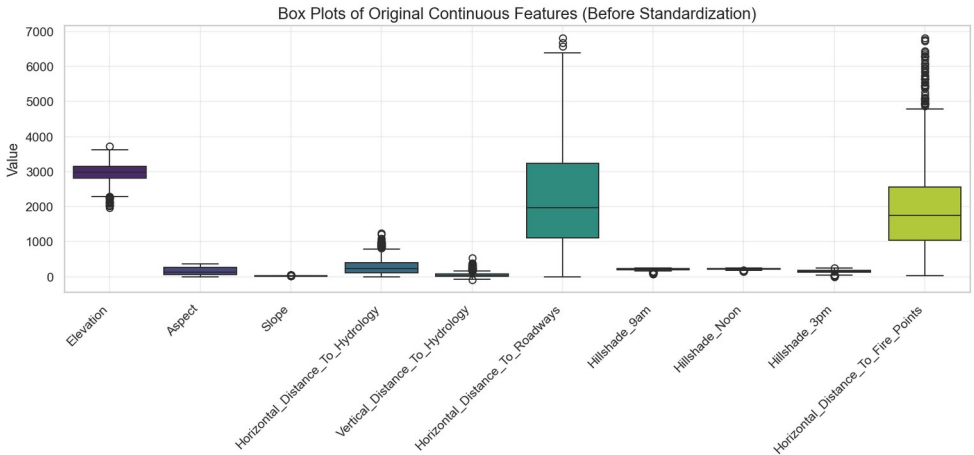
Category	Count	Representative Examples	Applied Preprocessing
Continuous	10	Elevation, Aspect, Slope, Hillshade (9am/noon/3pm), Distances to Hydrology/Road/Fire	StandardScaler (Mean=0, Std=1)

Binary indicators

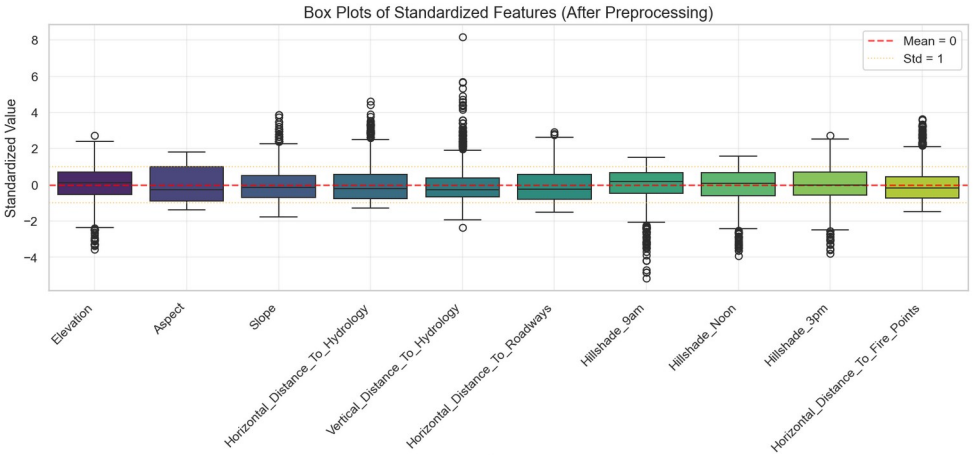
44 (4 wilderness, 40 soil)

Rawah, Neota, Comanche Peak, Cache la Poudre; Soil_Type_1–40

Keep as 0/1; mode-imputation ready; no scaling

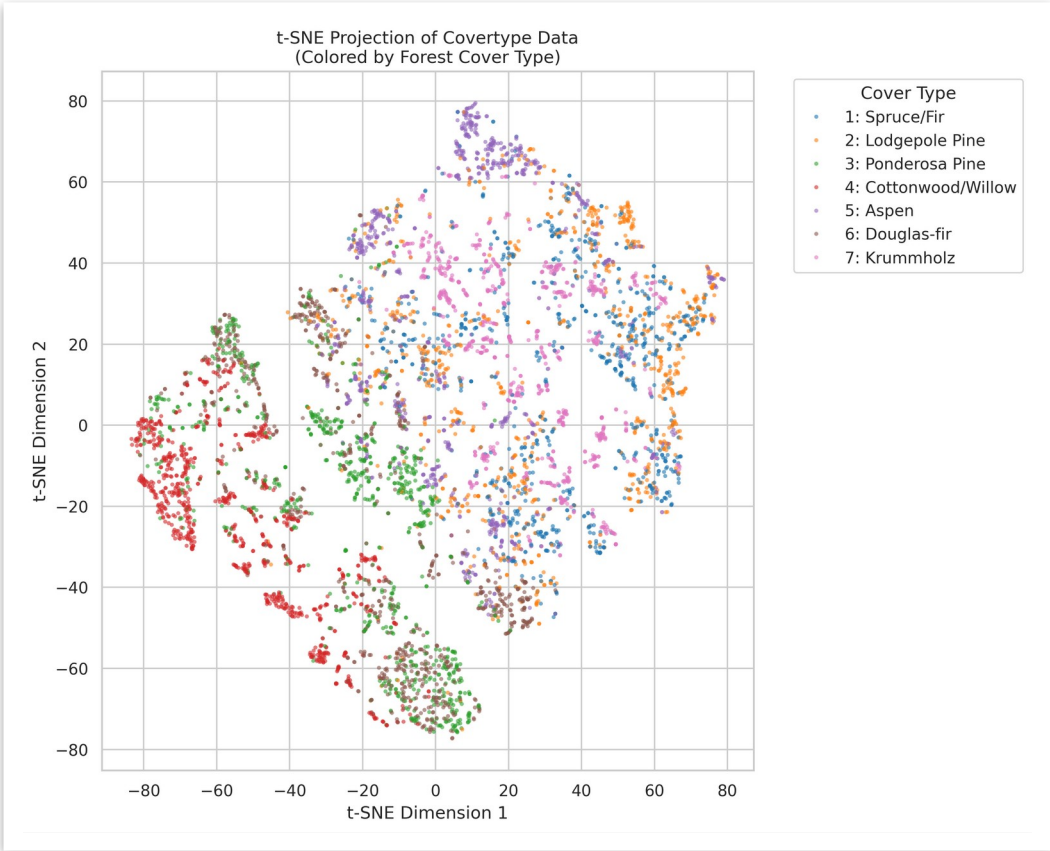


Preprocessing



03A t-SNE Data Visualization

Figure: t-SNE 2D Embedding of 7 Forest Cover Types



Key Observations & Interpretations

Local Structure, Not Clean Clusters — Classes 1 and 2 overlap across much of the embedding; prevalence does not imply separability.

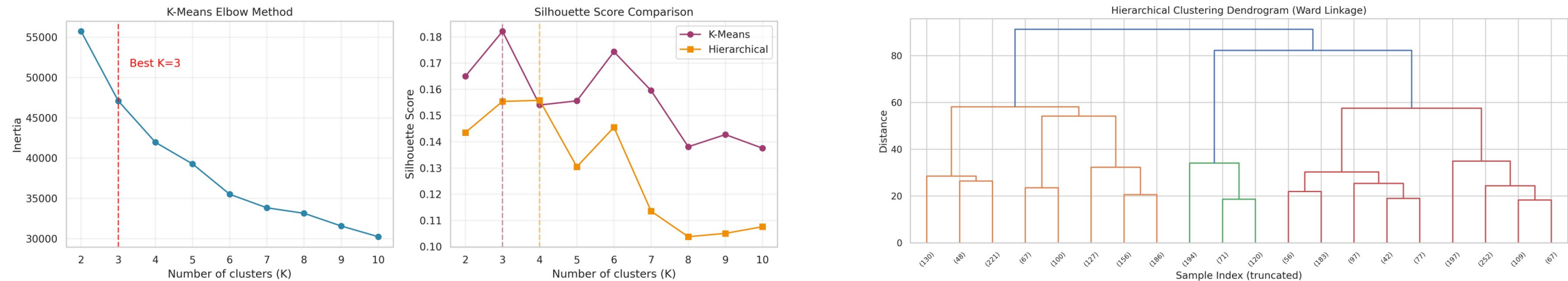
Rare-Class Pockets — Rare cover types form local pockets, but they are not globally isolated from the dominant classes.

High-Elevation Signal — Krummholz and some Spruce/Fir observations occupy more distinctive high-elevation regions.

Visualization Caveat — t-SNE preserves local neighborhoods—not global distances—and cannot justify a specific number of clusters.

5,999 stratified observations • continuous variables standardized • PCA 54 → 30 components • t-SNE used for visualization only

03B Clustering Analysis



Internal diagnostics favor K=3 for MiniBatchKMeans and K=4 for Ward; neither recovers the seven ecological labels.

MiniBatchKMeans

K=3 · silhouette=0.182 · CH=1317.241 · ARI=0.075

Ward Hierarchical

K=4 · silhouette=0.156 · CH=406.764 · ARI=0.103

Why Labels Are Not Recovered

Euclidean compactness on 10 continuous + 44 binary variables is not aligned with ecological class definitions.

Distance-Metric Check

Ward + Gower reaches ARI=0.215 at K=7—better, but still far below supervised Random Forest F1=0.908.

04 Predictive Model Training

Logistic Regression (Baseline)

Test Accuracy

0.676

Speed:

Fast training (linear complexity), low inference latency

Stability:

No overfitting ($\Delta F1 = 0.002$), very robust to noise

Macro F1 Score

0.660

Confusion Matrices - Logistic Regression

Train Set (n=68565)							
Actual \ Predicted	1	2	3	4	5	6	7
1	7973	2121	4	0	453	63	1385
2	2699	6299	173	3	2231	495	100
3	0	84	6674	1551	523	3167	0
4	0	0	127	1686	0	110	0
5	43	955	141	0	5226	280	0
6	0	99	2541	983	413	7963	0
7	1243	1	0	0	50	0	10706

Test Set (n=29385)							
Actual \ Predicted	1	2	3	4	5	6	7
1	3408	904	5	0	189	25	612
2	1175	2690	66	0	927	236	48
3	0	38	2852	654	234	1365	0
4	0	0	55	722	0	47	0
5	22	420	63	0	2216	127	0
6	0	40	1155	419	151	3378	0
7	536	0	0	0	20	0	4586

Full Set (n=97950)							
Actual \ Predicted	1	2	3	4	5	6	7
1	11381	3025	9	0	642	88	1997
2	3874	8989	239	3	3158	731	148
3	0	122	9526	2205	757	4532	0
4	0	0	182	2408	0	157	0
5	65	1375	204	0	7442	407	0
6	0	139	3696	1402	564	11341	0
7	1779	1	0	0	70	0	15292

Dataset Partitioning Strategy

Modeling sample: 97,950 stratified observations from the full 581,012. **Split:** 68,565 train / 29,385 held-out test (70/30).

Decision Tree (Regularized)

Test Accuracy

0.823

Macro F1 Score

0.819

Config: max_depth=24, min_samples_leaf=15 (fixed baseline)

Fit: Mild overfitting ($\Delta F1 = 0.046$), high predictive power

Confusion Matrices - Decision Tree

Train Set (n=68565)							
Actual \ Predicted	1	2	3	4	5	6	7
1	9863	1447	7	0	202	33	447
2	1487	9193	272	6	674	286	82
3	10	100	10291	555	120	923	0
4	0	0	22	1891	0	10	0
5	18	194	46	0	6339	48	0
6	18	68	886	355	65	10607	0
7	347	33	0	0	12	0	11608

Test Set (n=29385)							
Actual \ Predicted	1	2	3	4	5	6	7
1	3889	843	6	0	99	29	277
2	965	3451	135	3	386	160	42
3	2	59	4209	254	71	548	0
4	0	0	23	789	0	12	0
5	20	132	38	0	2634	24	0
6	11	43	533	161	41	4354	0
7	253	25	0	0	3	0	4861

Full Set (n=97950)							
Actual \ Predicted	1	2	3	4	5	6	7
1	13752	2290	13	0	301	62	724
2	2452	12644	407	9	1060	446	124
3	12	159	14500	809	191	1471	0
4	0	0	45	2680	0	22	0
5	38	326	84	0	8973	72	0
6	29	111	1419	516	106	14961	0
7	600	58	0	0	15	0	16469

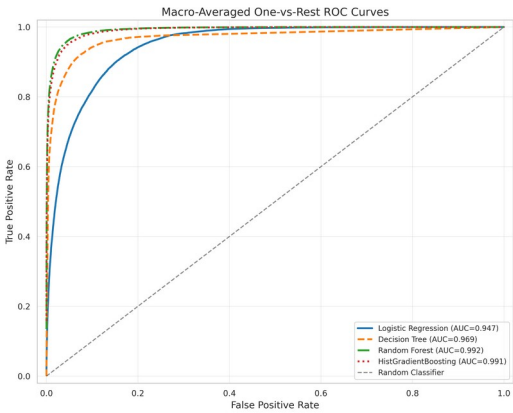
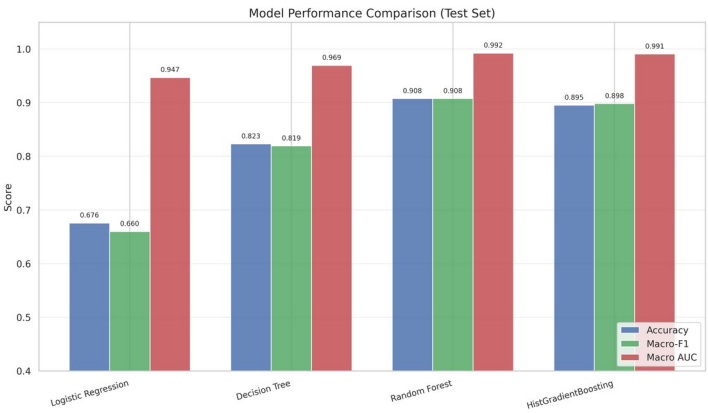
Class Imbalance Mitigation

Applied class-weighted training to adjust the loss function, assigning higher penalties to misclassifications of underrepresented classes. This technique improved minority class recall without significantly sacrificing overall precision.

05 Evaluation and Model Selection

Held-out Test and Validation Evidence

Model	Test Acc	Macro Prec.	Macro Recall	Macro-F1	95% CI	CV-F1	Macro AUC
Logistic Regression	.676	.646	.706	.660	[.654, .665]	.661±.003	.947
Decision Tree	.823	.803	.846	.819	[.812, .826]	.811±.004	.969
Random Forest	.908	.903	.913	.908	[.904, .911]	.902±.001	.992
HistGradientBoosting	.895	.896	.901	.898	[.894, .902]	.895±.003	.991



Selection: Random Forest has the best held-out macro-F1 (0.908) and AUC (0.992). The RF–HGB gap is stable under bootstrap but practically small ($\Delta F1=0.010$).

Statistical interpretation: McNemar $\chi^2=41.5$, $p<0.001$ shows different per-sample error patterns; it does not directly test the continuous macro-F1 difference.

06 Open-Ended Exploration

Beyond baseline models - deeper investigation of techniques and improvements

Feature Importance

Impurity and permutation rankings, followed by feature-group ablation.

Search Strategy

Grid search versus Bayesian optimization on the same 30k subset.

Feature-Group Ablation

Quantify the complementary value of terrain, wilderness, and soil indicators.

Calibration & Confidence

Brier score, ECE, Platt scaling, and confidence-based error review.

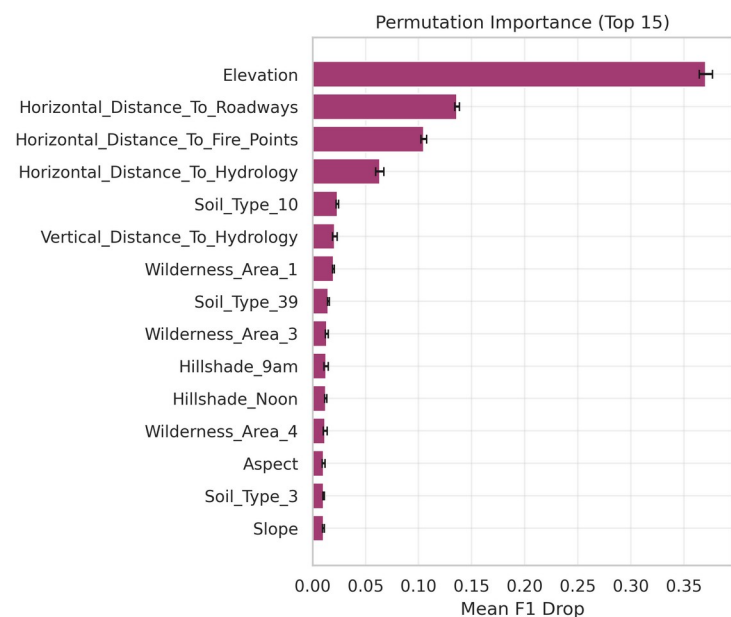
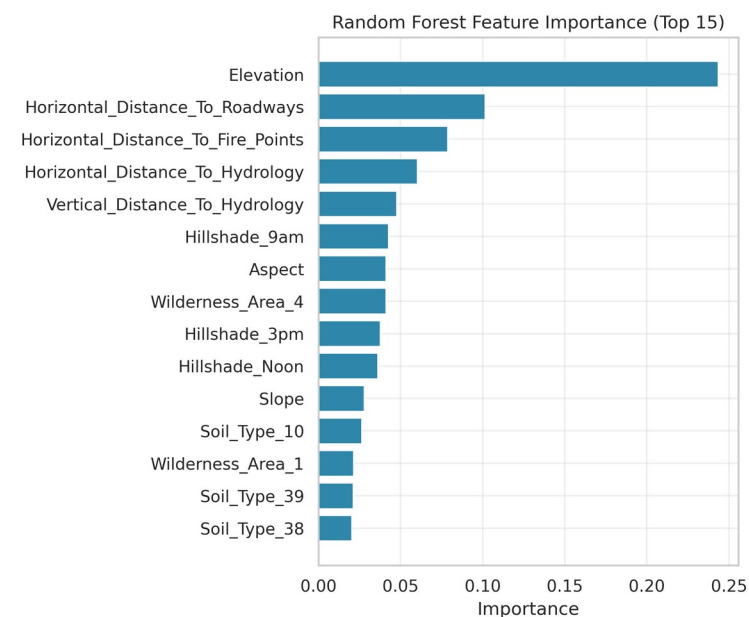
Robustness & Subgroups

Performance across classes, elevation bands, and wilderness areas.

Final Model Choice

Balance predictive performance, calibration, and subgroup reliability.

Feature Importance Analysis



Two independent importance methods agree on the leading predictors.

Interpretation: importance indicates predictive association, not a causal effect. Wilderness and soil indicators still add complementary context.

Impurity importance and held-out permutation importance provide complementary evidence.

01
Elevation

Impurity importance=0.243; permutation macro-F1 drop=0.371.

02 Road Distance

Second-largest permutation effect: macro-F1 drop=0.136.

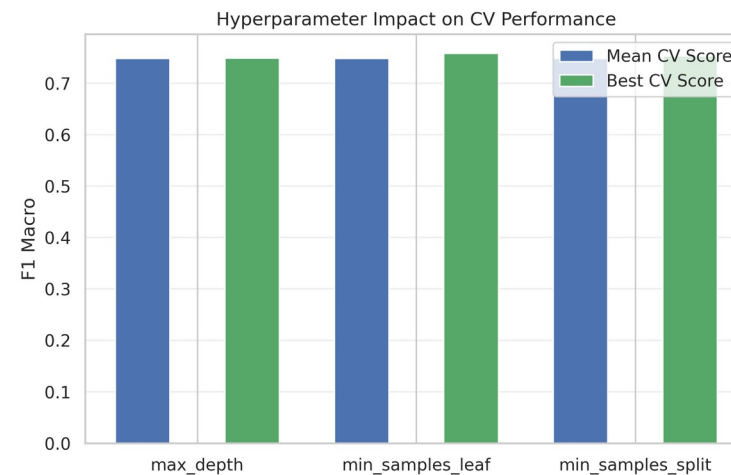
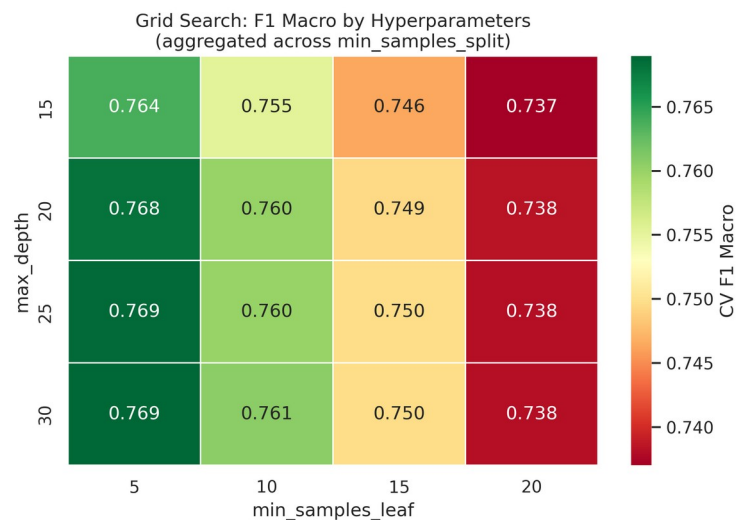
03 Fire-Point Distance

Third-largest permutation effect: macro-F1 drop=0.104.

Model Capacity

Terrain-only RF=0.874; top-5 RF=0.877; full RF=0.908. LR polynomial features remain 0.584.

Hyperparameter Search Strategy



Decision Tree Grid

48 configurations, 3-fold CV. Best CV-F1=0.769; held-out test F1=0.797 versus 0.819 baseline ($\Delta=-0.022$).

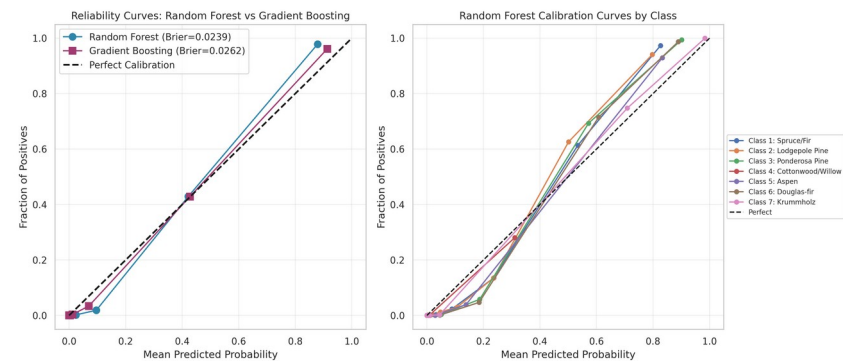
Ensemble Grid

RF 0.908 $\rightarrow$ 0.886 and HGB 0.898 $\rightarrow$ 0.889.
Increasing RF trees 120 $\rightarrow$ 200 adds only +0.001 F1.

Optuna on Same 30k

100 TPE trials recover test F1=0.823: +0.026 over grid and +0.003 over baseline, within the baseline CI. Search strategy—not subset size—is the key lesson.

Model Calibration Analysis



Baseline Random Forest
0.024

Brier score (higher = poorer)

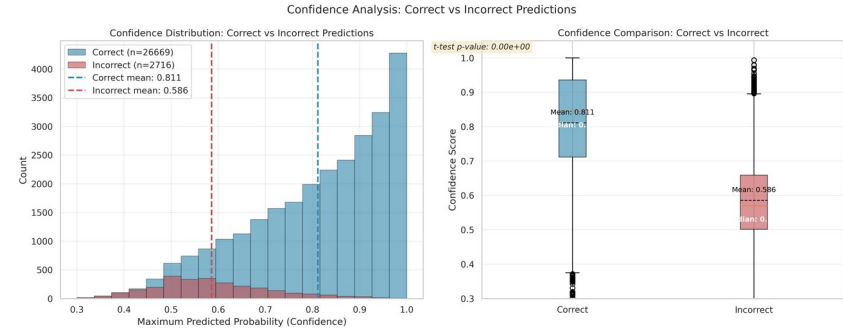
RF ECE
0.125

Under-confident: 0.783 conf vs 0.908 acc

HGB ECE
0.041

Best calibrated

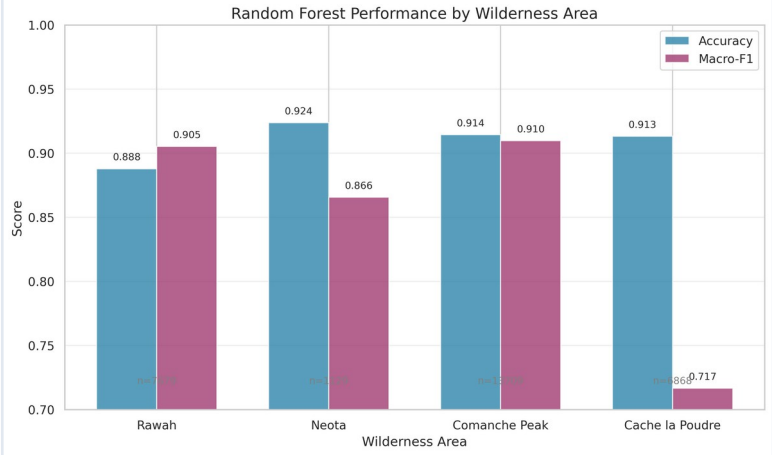
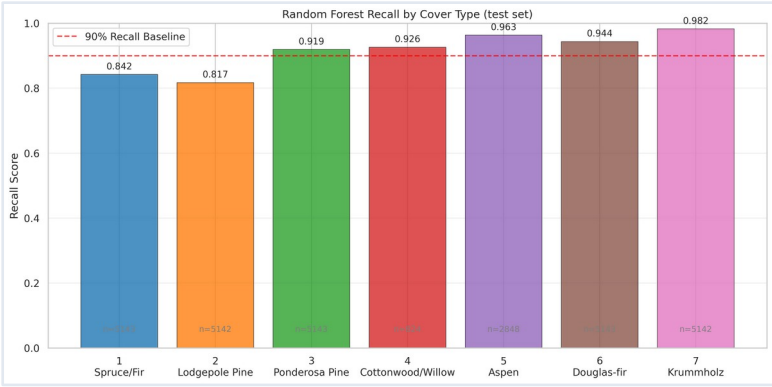
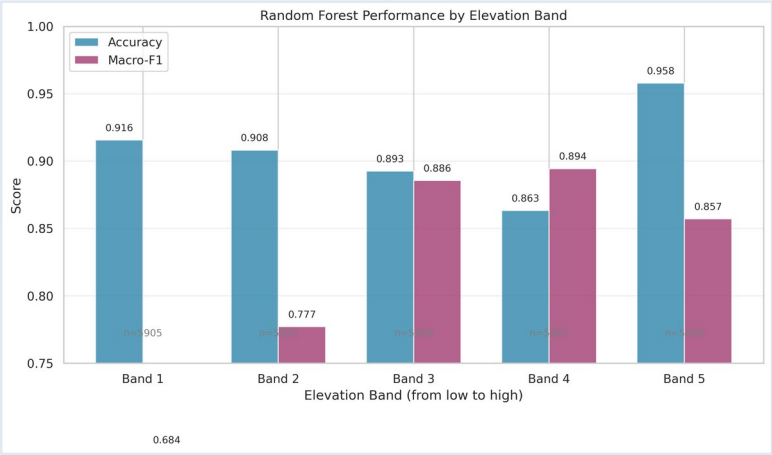
Calibration: RF is under-confident (ECE=0.125; mean top-label confidence=0.783 vs accuracy=0.908). HGB is best calibrated (ECE=0.041), while Platt scaling reduces RF Brier 0.024 → 0.021.
McNemar: $\chi^2=41.5$, $p<0.001$ indicates different per-sample error patterns—not a direct macro-F1 significance test.



Robustness range: elevation-band accuracy 0.863–0.958 and macro-F1 0.684–0.900; the most diverse band is hardest. Cache la Poudre has macro-F1=0.716.

Confidence separates many errors: correct predictions average 0.811 confidence; incorrect predictions average 0.586.

Robustness and Subgroup Analysis



Terrain Bands

Accuracy ranges 0.863–0.958 and macro-F1 0.684–0.900. The most ecologically diverse elevation band is hardest; high-elevation Krummholz regions are easier.

Class-Level Recall

Lodgepole Pine is lowest at 0.817, followed by Spruce/Fir at 0.842. Aspen, Douglas-fir, and Krummholz each exceed 0.94 recall.

Wilderness Areas

Macro-F1: Rawah 0.906, Neota 0.866, Comanche Peak 0.910, Cache la Poudre 0.716. Geographic composition drives concentrated error patterns.

07 Conclusion and Future Outlook

Key Conclusions

Best Predictive Model

Random Forest: accuracy=0.908, macro-F1=0.908, macro AUC=0.992. Its 0.010 F1 lead over HGB is stable but practically small.

Natural Clusters ≠ Ecological Labels

K-Means ARI=0.075, Ward ARI=0.103; Gower improves Ward to 0.215 but remains far below supervised performance.

Topographic Factors Drive Predictions

Elevation is the most critical predictor; topographic features decisively shape classification results.

Evidence and Limitations

- Modeling sample covers ≈17% of 581,012 observations
- RF–HGB gap is stable under bootstrap, but practically small
- Grid search degrades all three models; Optuna recovers the DT baseline
- Reference full-data benchmarks reach roughly 0.95–0.96 accuracy
- Errors remain concentrated by class, elevation, and wilderness area

Future Work Outlook

Train on the full dataset

Scale the validated pipeline from the 97,950-row modeling sample to all 581,012 observations and compare with published benchmarks.

Tune ensembles with Bayesian search

Extend Optuna from Decision Tree to Random Forest and HGB, using nested validation and early pruning.

Calibrate and abstain

Evaluate calibrated probabilities and confidence-based abstention when uncertain predictions require review.

Validate subgroups explicitly

Track class-, elevation-, and wilderness-specific metrics rather than relying on a single aggregate score.

A misty forest scene with sunlight filtering through the trees. The sun is positioned in the upper right, creating a bright glow and long, soft rays of light that penetrate the mist. The trees are tall and slender, their trunks dark against the lighter, hazy background. The ground is covered in fallen leaves and branches, adding to the sense of a quiet, natural setting.

Q&A

Thank You

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